Unifying Short-Timescale Affect Dynamics and Long-Timescale Memory Consolidation in Large-Scale Multi-Agent Social Simulations

Executive Summary

The structural unification of a short-timescale affective state (mood) with a long-timescale memory system (power-law forgetting) is an architectural necessity for producing biologically plausible agents in large-scale social simulations. Relying on Large Language Models (LLMs) to infer affective states from raw text logs produces unacceptable persona drift, lacks temporal continuity, and fails to support arbitrary simulation time-skips. The analysis evaluates the integration of these two temporal domains within the constraints of lazy evaluation and event-driven architectures.

The evaluation of the proposed design hypotheses (H1-H8) yields the following determinations, utilizing evidence from affective chronometry, psychometrics, and computational cognitive modeling.

Hypothesis	Status	Justification
H1. Intensity-dependent recovery	Confirmed	Empirical affective chronometry demonstrates that initial emotional intensity strictly predicts emotion duration and alters the hazard rate of decay.
H2. Sum of decaying impulses	Modified	While continuous autoregressive models are standard, a heavy-tailed decaying impulse summation is mathematically required to support exact closed-form lazy evaluation without background ticks.
H3. Personality-dependent	Confirmed	Emotional inertia (resistance to affective

recovery		change) is highly correlated with Neuroticism and maladjustment, acting as a distinct parameter from initial reactivity.
H4. Mood-congruent retrieval	Confirmed	Meta-analyses establish mood-congruent memory as a powerful retrieval bias, distinct from the structural degradation of the memory trace itself.
H5. Arousal-driven encoding	Confirmed	Amygdala-driven modulation of memory consolidation dictates that physiological arousal scales the initial stability of episodic memory traces.
H6. Retrieval triggers mood	Confirmed	The reactivation of salient memories serves as an internal event trigger, extending emotion duration through rumination pathways.
H7. Memory-driven appraisal	Confirmed	Appraisal theories position memory-driven relevance and coping potential as the primary antecedents to subsequent emotion elicitation.
H8. Shared decay family	Modified	Both mechanisms require heavy-tailed decay, but memory conforms to the FSRS specific power-law, whereas mood requires bounded, saturating functions to prevent

		divergence.

Evidence Review

Q1. Temporal Dynamics of Affect

(Confidence: Strong)

The temporal unfolding of emotions, formally termed affective chronometry, contradicts the assumption that emotional states decay exponentially back to a baseline. Discrete-time survival analyses of emotion duration reveal that the hazard rate of emotions—defined as the conditional probability that an emotional episode will end in a given time interval—is highly variable across emotion types and durations[1]. The hazard rate of fear drops steeply and linearly, resulting in a short median lifetime of approximately 16 minutes[1]. Conversely, emotions such as anger and joy exhibit a distinct hazard function that rises slightly before undergoing a steep drop after one hour, yielding longer median lifetimes of 22 and 26 minutes, respectively[1]. Crucially, the survivor functions indicate a heavy-tailed distribution. Emotions that survive the initial hourly intervals are disproportionately likely to endure for significantly longer periods[1]. This heavy-tailed nature implies that affect recovery is mathematically better modeled by power-law dynamics rather than the memoryless exponential decay commonly implemented in naive computational agents. Furthermore, the duration of an emotional episode is heavily modulated by the initial intensity of the event and the presence of rumination[1].

In computational psychology, continuous-time affective modeling frequently relies on the Ornstein-Uhlenbeck (OU) process, which is the continuous analogue of a first-order autoregressive AR(1) model[3]. While AR(1) models successfully capture the tendency of a system to revert to a mean baseline, the heavy-tailed real-world data and the compounding effect of rumination imply that acute emotional impulses persist longer than a standard exponential AR(1) process predicts. To capture both the mean-reverting nature of the OU process and the heavy-tailed survival data from affective chronometry, the computational model must utilize dynamic, intensity-dependent time constants governed by power-law impulse formulations.

Q2. Individual Differences

(Confidence: Strong)

Personality traits systematically govern affective dynamics. The Cybernetic Big Five Theory (CB5T) provides a mechanistic framework, positioning traits as parameters of evolved cybernetic mechanisms regulating entropy and goal-directed behavior[6]. Within CB5T, the meta-trait of Stability (representing the shared variance of Conscientiousness, Agreeableness, and inverse Neuroticism) governs the protection of existing behavioral policies from disruption, while Plasticity (Extraversion and Openness) governs exploration[7].

Empirical research on emotional inertia—defined as the degree to which emotional states resist change and carry over from one moment to the next—demonstrates a profound correlation with Neuroticism and psychological maladjustment[9]. High Neuroticism correlates with high

emotional inertia, meaning affective states persist longer and are more self-predictive over time. This inertia is independent of the initial reactivity to a stimulus[9]. Individuals with high Extraversion exhibit heightened reactivity (initial response amplitude) to positive stimuli, while highly neurotic individuals exhibit heightened reactivity to negative stimuli and systemic uncertainty[12]. Consequently, reactivity (the amplitude of the emotional impulse) and inertia (the decay rate of the impulse) are distinct, separable mathematical parameters, though both are heavily modulated by the agent's Big Five profile. Furthermore, the cybernetic framework supports co-fluctuation under stress, where acute adversity increases entropy, driving Neuroticism up while suppressing Extraversion and Conscientiousness[6].

Q3. Mood and Memory Interaction

(Confidence: Strong for Mood-Congruent and Arousal; Contested for Mood-Dependent)

The interaction between affect and memory operates primarily through two independent, neurobiologically distinct pathways: encoding enhancement and retrieval bias. Regarding encoding, extensive neurobiological literature establishes that emotional arousal activates the basolateral amygdala via noradrenergic pathways, which subsequently modulates memory consolidation in regions such as the hippocampus[13]. This consolidation modulation mechanism implies that the initial structural stability of a memory trace is a direct mathematical function of the organism's physiological arousal at the exact time of encoding or shortly thereafter. Experiences accompanied by significant adrenal stress hormone release are consolidated into more durable traces[16].

Regarding retrieval, a strict distinction must be drawn between mood-dependent memory and mood-congruent memory. Mood-dependent memory (the hypothesis that recall is superior if the retrieval mood matches the encoding mood, regardless of the memory's intrinsic valence) has repeatedly failed rigorous replication and proved highly unreliable in laboratory settings[18]. It represents folklore that should not be encoded into the architecture. Conversely, mood-congruent memory (the hypothesis that current mood biases the retrieval of memories matching that specific mood valence) is a highly robust phenomenon. Meta-analyses demonstrate a baseline positive bias ($d = 0.15$) in healthy individuals, which severely inverts to a negative bias ($d = -0.19$) in clinically depressed individuals[18]. This bias operates as a transient filter in healthy individuals but locks into place in maladaptive states. The depressed brain does not merely retrieve negative memories; it actively constructs them, filling gaps with mood-congruent material[18]. Therefore, current mood must act as a continuous modifier on the retrievability of similarly valenced memories during the querying phase, without altering the underlying consolidated stability.

Q4. Appraisal and Construal

(Confidence: Moderate)

Appraisal theories dictate that emotions are not elicited directly by raw stimuli, but by the subjective appraisal of the stimulus regarding its relevance, implications, and coping potential. Scherer's Component Process Model positions this appraisal as a sequential, memory-dependent operation[20]. The construal of a situation relies entirely on the retrieval of

episodic precedents to estimate predictability, threat, and goal-congruence.
While the DIAMONDS framework successfully categorizes situations into eight discrete dimensions (Duty, Intellect, Adversity, Mating, pOsitivity, Negativity, Deception, Sociality)[22], driving cybernetic mood dynamics exclusively through discrete categories results in a loss of continuous valuation. A continuous appraisal dimension representation—specifically mapping Valence, Arousal, and Controllability—aligns more accurately with the cybernetic parameters of CB5T. To operationalize appraisal, the event classification must query the agent's memory for relevant past experiences; if past betrayals are retrieved during a neutral event, the appraised threat level increases, directly modulating the intensity of the resulting emotional impulse.

Q5. Computational Models

(Confidence: Moderate)
Existing formal models of affect dynamics provide frameworks for coupling cognition and emotion. Reinforcement-learning models, notably the "mood-as-momentum" model proposed by Eldar and Niv, conceptualize mood as an aggregate of recent reward prediction errors[25]. In these models, positive prediction errors elevate mood, which in turn acts as a ubiquitous bias on future expected values. This bidirectional coupling creates a mathematical feedback loop where mood alters environmental perception, and perception iteratively alters mood.
Cognitive architectures such as ACT-R and affective frameworks like FAtiMA implement PAD-based (Pleasure, Arousal, Dominance) emotional states, typically utilizing linear or standard exponential decay functions. However, none of these standard architectures inherently account for the power-law forgetting curves observed in human memory consolidation, nor are they optimized for the lazy evaluation required by asynchronous, multi-agent LLM simulations. The proposed design must therefore adapt the bidirectional feedback loop of the mood-as-momentum model while replacing exponential decay with heavy-tailed power-law decay.

Q6. LLM-Agent Literature

(Confidence: Moderate)
The simulation of society using LLMs, exemplified by Generative Agents, relies heavily on text-based memory streams, spatial distance heuristics, and periodic reflective consolidation[27]. While these agents exhibit plausible emergent social behaviors, their emotional states are typically tracked via prompt engineering rather than underlying mathematical formalisms. Relying purely on an LLM to infer its mood from a text log of retrieved memories leads to severe persona drift and a total lack of emotional inertia. The LLM has no temporal decay function native to its context window; a memory from ten years ago and a memory from ten seconds ago carry identical affective weight unless explicitly timestamp-engineered.
Introducing a quantitative, cybernetic mood tracker external to the LLM stabilizes the agent's baseline and ensures biologically plausible affective chronometry. Empirical validation of LLM-simulated affect indicates that while LLMs can accurately classify discrete emotions, they fail to natively simulate the temporal decay of those emotions over simulated time-skips. Grounding the LLM's generative output in strict mathematical constraints provided by an external state vector ensures realistic behavior.

Q7. Lazy and Time-Skip Compatibility

(Confidence: Strong)
The architectural requirement for lazy evaluation—computing the current state only at query time without continuous background clock ticks—necessitates strictly closed-form mathematical functions. The FSRS (Free Spaced Repetition Scheduler) algorithm achieves this for memory via a power-law retrievability function:

$$R(t, S) = \left(1 + \text{FACTOR} \cdot \frac{t}{S}\right)^{\text{DECAY}}$$

Specifically, FSRS-4.5 utilizes $R(t, S) = (1 + \frac{19}{81}\frac{t}{S})^{-0.5}$ [29].
To align mood with lazy evaluation, mood cannot be a tick-based autoregressive update. It must be modeled as a sum of historical mood impulses. Given that human emotion hazard rates drop and then flatten out[1], an exponential decay $e^{-t/\tau}$ approaches zero too rapidly at large t. A power-law impulse function $\Delta M \cdot (1 + t/\tau)^{-\alpha}$ allows for an exact, closed-form summation at any arbitrary time t.
Lazy evaluation is mathematically exact for the mood state itself. However, the coupling—where mood influences memory encoding in the background—requires an approximation. Because the system is event-driven, the mood is only evaluated at the exact timestamp of an event. A time-skip of 40 simulated years implies $\Delta t \to \infty$. Under a power-law formulation, the summation mathematically decays to zero, and the agent's mood gracefully regresses to its static personality baseline without requiring computationally expensive background tick updates.

Q8. Failure Modes and Safeguards

(Confidence: Strong)
Coupled dynamical systems of mood and memory are highly susceptible to runaway feedback loops. If an agent retrieves a sad memory, it generates a negative affective impulse; this negative mood biases subsequent retrievals toward further sad memories, infinitely amplifying the state. While this reflects the clinical reality of depressive rumination[18], it leads to mathematical divergence (saturation) in a simulation.
Safeguards require specific boundary conditions grounded in cybernetic theory. First, a bounding function, such as the hyperbolic tangent ($\tanh$), must strictly limit absolute mood. Second, an adaptation mechanism must be introduced where repeated retrieval of the exact same memory yields diminishing affective impulses, simulating neurological habituation[31]. Third, an integration ceiling or affective refractory period is required to prevent infinite loops during highly active event clusters.

Q9. Classifier Reliability

(Confidence: Moderate)
Utilizing an LLM to classify every micro-event into eight DIAMONDS dimensions is computationally expensive and highly prone to categorization hallucinations. While DIAMONDS

maps robustly to behavioral outcomes in human experience sampling[23], reducing event classification to continuous embedding-based classifiers or simpler Valence-Arousal regressions is computationally cheaper and mathematically more reliable. If LLM classification is maintained, constraining the output to a normalized dimensional vector $[0, 1]$ via strictly typed output formats (e.g., JSON schema enforcement) minimizes parser failures and ensures the mood equations receive bounded, predictable inputs. A continuous appraisal dimension representation (Valence, Arousal, Controllability) is vastly superior for driving cybernetic mood dynamics compared to discrete categorical tags.

Q10. Validation

(Confidence: Strong)

Validation of the unified system must be grounded in Whole Trait Theory (WTT), which posits that personality traits are essentially the means of density distributions of momentary states, with approximately 80% of total behavioral variance occurring within-person across situations[32]. The simulation is successfully validated if, over thousands of varied events, the density distribution of a specific agent's mood states matches the expected mean and standard deviation of their assigned Big Five profile. Furthermore, the autocorrelation of the agent's simulated mood series must match human empirical inertia bounds—typically $r = 0.2$ to 0.6 at lag-1 depending on the agent's Neuroticism parameter[9].

The Unified Formulation

The system operates via two interacting layers: the long-timescale FSRS Memory layer and the short-timescale Mood layer. Both utilize closed-form equations to guarantee compatibility with event-driven architectures and arbitrary time-skips.

Core Equations and Lazy Evaluation

Memory Retrievability (FSRS-4.5 Based) The retrievability R of memory node i, evaluated at current time t, is a closed-form power-law derived from the DSR (Difficulty, Stability, Retrievability) model[29].

$$R_i(t) = \left(1 + \frac{19}{81}\frac{t - t_{last_retrieved}}{S_i}\right)^{-0.5}$$

where S_i is the memory's structural stability, representing the time required for retrievability to drop to 90%.

Mood State (Closed-Form Lazy Summation)

To compute mood at time t without tick updates, the state is the sum of a baseline and all historical impulses, passed through a hyperbolic tangent saturation function to prevent divergence.

$$M(t) = \tanh\left(M_{base} + \sum_{j \in E} I_j \cdot v_j \cdot \left(1 + \frac{t - t_j}{\tau_j}\right)^{-\alpha} \right)$$

- M_{base}: Trait baseline mood (derived from Extraversion and Neuroticism).
- I_j: Intensity of event j (influenced by event severity and agent reactivity).
- v_j: Valence of event j ($\in [-1, 1]$).
- τ_j: Emotion duration parameter (mood stability), determined by agent inertia and event importance[1].
- α: Global decay shape parameter. Setting $\alpha = 0.5$ aligns the affective hazard rate with the memory decay shape, producing the heavy-tailed survival function observed in empirical chronometry[1].

The Memory-Mood Coupling

The interaction between the systems occurs via three mathematically defined pathways.

1. Mood to Memory Encoding (Arousal-Modulated Consolidation) Upon encoding a new episodic memory at time t_{enc}, the initial FSRS stability S_0 is scaled by the absolute intensity of the agent's current mood, which acts as a proxy for physiological arousal[15].

$$S_{0,new} = S_{0,base} \cdot (1 + \lambda_{enc} \cdot |M(t_{enc})|)$$

This ensures that events occurring during highly emotional states are consolidated into more durable traces.

2. Mood to Memory Retrieval (Mood-Congruent Bias) When querying memory, the objective retrievability R_i is modified by the agent's current mood to produce a final contextual retrieval score. This implements the meta-analytic findings of Matt & Vazquez[18] without altering the underlying FSRS structure.

$$\text{Score}_i(t) = R_i(t) + \beta_{mc} \cdot (v_i \times M(t))$$

If the memory valence v_i matches the current mood $M(t)$, the retrieval score increases, creating a transient filter that favors mood-congruent recall.

3. Memory to Mood (Rumination) When a highly salient memory i is retrieved, it acts as an internal event, generating a new mood impulse at time t. The intensity of this impulse is a fraction of the memory's original arousal, scaled by the agent's rumination trait[2].

$$I_{rumination} = \gamma_{rum} \cdot A_i \cdot R_i(t)$$

Personality Parameter Sampling

Trait parameters are sampled directly from the agent's Big Five profile to individualize the cybernetic equations.

Trait / Facet	Modulated Parameter	Direction	Evidence Strength
Neuroticism	Emotional Inertia (τ)	Positive	Strong[9]
Neuroticism	Rumination coefficient (γ_{rum})	Positive	Strong[2]
Neuroticism	Reactivity to Negative Events	Positive	Strong[7]
Extraversion	Reactivity to Positive Events	Positive	Strong[7]
Extraversion	Baseline Mood (M_{base})	Positive	Moderate
Conscientiousness	Mood-Congruent Bias (β_{mc})	Negative (Damping)	Weak
Openness	Appraisal Complexity	Positive	Unverified

Graph Schema Changes

To support this unified architecture, the Neo4j/Graphiti-style database requires specific modifications. Storing individual mood impulses as graph nodes allows the system to seamlessly cull decayed impulses during standard graph maintenance operations.

```cypher
// New Node: MoodImpulse (linked to the Agent)
CREATE (m:MoodImpulse {
  intensity: Float,     // I_j (Amplitude of the event)
  valence: Float,       // v_j (Range: -1.0 to 1.0)
  tau: Float,           // tau_j (Inertia/Stability of the impulse)
  timestamp: Integer,   // t_j (Simulation tick)
  event_ref: String     // Optional UUID link to triggering event/memory
})

// Modified Node: Memory (Adding valence and arousal for coupling)
CREATE (mem:Memory {
  stability: Float,     // S parameter in FSRS
  difficulty: Float,    // D parameter in FSRS
  valence: Float,       // Event valence [-1.0, 1.0]
  arousal: Float,       // Event arousal [0.0, 1.0]
  valid_at: Integer,    // Encoding timestamp
  last_retrieved: Integer // Updates upon successful recall
})

// Edge: Agent to MoodImpulse
CREATE (a:Agent)-[:EXPERIENCED]->(m:MoodImpulse)
```

Optimization Note: During simulation runtime, a background maintenance job must delete MoodImpulse nodes where $(1 + \frac{t-t_j}{\tau_j})^{-\alpha} < \epsilon$ (where ϵ is a near-zero threshold) to prevent the agent's impulse history from expanding infinitely.

Python Pseudocode

The following pseudocode outlines the implementation of the lazy evaluation and coupled dynamics. It is designed for integration into event-driven simulation environments.

```python
import math
from typing import List

class MoodImpulse:
    def __init__(self, intensity: float, valence: float, tau: float, timestamp: float):
        self.intensity = intensity
```

```
        self.valence = valence
        self.tau = tau
        self.timestamp = timestamp

class Agent:
    def __init__(self, n_score: float, e_score: float):
        self.neuroticism = n_score
        self.extraversion = e_score
        self.impulses: List[MoodImpulse] = []

        # Cybernetic trait mappings
        self.m_base = (e_score - 0.5) * 0.5
        self.inertia_mult = 1.0 + (n_score * 2.0)
        self.reactivity_pos = 1.0 + (e_score * 1.5)
        self.reactivity_neg = 1.0 + (n_score * 1.5)
        self.rumination_mult = n_score

    def compute_mood_lazily(self, current_time: float) -> float:
        """
        Computes current mood exactly at query time (no ticks).
        Utilizes power-law summation and tanh bounding.
        """
        decay_shape_alpha = 0.5
        active_sum = 0.0
        alive_impulses = []

        for imp in self.impulses:
            delta_t = current_time - imp.timestamp
            if delta_t < 0:
                continue

            # Power-law decay mimicking heavy-tail affective chronometry
            decay_factor = math.pow(1.0 + (delta_t / imp.tau), -decay_shape_alpha)
            impulse_val = imp.intensity * imp.valence * decay_factor

            # Culling threshold to prevent array growth
            if abs(impulse_val) > 0.01:
                active_sum += impulse_val
                alive_impulses.append(imp)

        self.impulses = alive_impulses # State cleanup

        # Tanh bounds the mood to [-1, 1] to prevent runaway divergence
```

```
        return math.tanh(self.m_base + active_sum)

    def process_event(self, event_valence: float, event_intensity: float, current_time: float):
        """Creates a new mood impulse based on an external event."""
        # Modulate reactivity by traits
        if event_valence > 0:
            adjusted_intensity = event_intensity * self.reactivity_pos
        else:
            adjusted_intensity = event_intensity * self.reactivity_neg

        # Tau (stability) determined by event intensity and agent inertia
        tau = max(1.0, adjusted_intensity * 100.0 * self.inertia_mult)

        new_impulse = MoodImpulse(
            intensity=adjusted_intensity,
            valence=event_valence,
            tau=tau,
            timestamp=current_time
        )
        self.impulses.append(new_impulse)

def score_memory_retrieval(agent: Agent, memory, current_time: float) -> float:
    """
    Scores memory retrievability using FSRS and mood-congruent bias.
    Memory object expects attributes: stability, last_retrieved, valence
    """
    delta_t = current_time - memory.last_retrieved

    # FSRS 4.5 baseline retrievability
    retrievability = math.pow(1.0 + (19.0/81.0) * (delta_t / memory.stability), -0.5)

    current_mood = agent.compute_mood_lazily(current_time)

    # Mood-congruent memory bias (Matt & Vazquez, 1992)
    mood_bias_weight = 0.2
    bias = mood_bias_weight * (current_mood * memory.valence)

    return max(0.0, min(1.0, retrievability + bias))

def retrieve_and_ruminate(agent: Agent, memory, current_time: float):
    """Retrieving a memory creates a rumination impulse."""
    ret_score = score_memory_retrieval(agent, memory, current_time)
    if ret_score > 0.5: # Threshold for successful retrieval
```

```
        rumination_intensity = memory.arousal * agent.rumination_mult * 0.5
        agent.process_event(
            event_valence=memory.valence,
            event_intensity=rumination_intensity,
            current_time=current_time
        )
        memory.last_retrieved = current_time
```

Safeguards and Failure Modes

Because the system implements mathematically rigorous feedback loops (memory affects mood via rumination; mood affects memory via congruent retrieval), dynamical instability is a severe risk in isolated multi-agent systems.

Failure Mode	Detection	Mitigation / Safeguard	Literature Grounding
Depressive Attractor	Agent repeatedly recalls negative memories, generating negative rumination impulses, trapping mood at -1.0 permanently.	Implement $\tanh$ bounding function on total mood. Integrate an affective refractory period (impulses from rumination capped per hour).	Clinical depression features runaway mood-congruent memory[18], but healthy brains utilize inhibitory control.
Array Explosion	Agent impulses array grows infinitely over simulation steps, causing memory leaks.	Memory cleanup loop: strictly cull impulses when $I \cdot (1 + t/\tau)^{-\alpha} <$.	Emulates absolute affective forgetting and baseline return.
Mood Saturation	Extreme external events stack up rapidly, pegging mood to 1.0 instantly.	Non-linear impulse addition or $\tanh$ saturation prevents linear combinations from breaking logic bounds.	Corresponds to the Weber-Fechner law of diminishing psychophysical sensitivity.

Time-Skip Errors	Skipping 40 simulation years results in division by zero or infinite retrievability.	Strict type bounding in Python. The power-law formulation safely evaluates to $\approx$ as $t \to$.	Asymptotically safe mathematical limits.

Validation Plan

To scientifically validate that the simulated agents exhibit realistic psychological behavior, the simulation outputs must be measured against established psychometric baselines:

1. **Whole Trait Theory (WTT) Variance Check**: Run the simulation for 30 in-game days. Measure the mood of agents at random intervals. The resulting density distributions must demonstrate that within-person variance accounts for approximately 50% to 80% of total variance, directly matching Fleeson's experience sampling findings[32].
2. **Emotional Inertia Targets**: Calculate the lag-1 autocorrelation of the mood time series. Agents possessing high Neuroticism must display an autocorrelation of $r \approx 0.5 \text{ to } 0.6$, while low Neuroticism agents must drop to $r \approx 0.2$, matching Kuppens' clinical baseline parameters[9].
3. **Mood-Congruent Memory Ablation**: Measure the ratio of positive to negative memories retrieved during simulation queries. A healthy agent in a neutral state should exhibit a slight positive bias ($d \approx 0.15$). When the agent is mathematically forced into a negative mood state, the retrieval ratio must invert to favor negative memories ($d \approx -0.19$)[18].
4. **Arousal-Consolidation Check**: Compare the FSRS stability (S) of memories encoded during a neutral state versus an extreme mood state. The S values for high-arousal memories should statistically survive longer retention intervals, confirming the operationalization of the McGaugh amygdala mechanism[15].

Open Problems

The literature and this formulation leave specific computational parameters unresolved, which must be calibrated empirically during simulation tuning:

The optimal power-law shape parameter (α) for affective hazard rates is not mathematically standardized in the literature. While memory decay in FSRS-4.5 utilizes $\alpha = 0.5$ [29], applying this directly to mood impulses may cause emotions to last artificially long in highly volatile environments. Empirical tuning is required to align the parameter with Verduyn's median emotion lifetimes.

Furthermore, using LLMs to continuously determine event valence and intensity introduces latency and LLM bias. It remains entirely untested whether an LLM can consistently output

normalized continuous valence/intensity values across thousands of diverse edge cases without artificially drifting toward a "neutral" safe baseline due to RLHF safety training.
Finally, cross-trait interactions remain theoretically complex. The CB5T literature maps traits to broad cybernetic parameters, but precisely how Conscientiousness dampens emotional impulses (regulatory control) versus how Agreeableness alters the valence of social events is theoretically sound but computationally unquantified in current open-source agent frameworks.

Literature Annotations

The foundational evidence dictating this architecture is sourced directly from primary psychological and computational literature.

Source Identification	Relevance to System Architecture
Verduyn, P. et al. (2012). *Emotion*.[1]	Demonstrates that emotion duration is highly variable, relies on event importance and rumination, and that the hazard function of emotions is non-linear, requiring heavy-tailed decay over simple exponential functions.
Kuppens, P. et al. (2010). *Psychological Science*.[9]	Establishes the concept of emotional inertia (resistance to affective change) and links it strongly to psychological maladjustment and Neuroticism, dictating the decay parameter τ.
DeYoung, C. G. (2015). *Journal of Research in Personality*.[6]	Provides the Cybernetic Big Five Theory (CB5T) framework linking Big Five traits (Stability, Plasticity) to dynamic cybernetic responses like entropy regulation and reward sensitivity.
Matt, G. E., Vazquez, C., et al. (1992). *Clinical Psychology Review*.[18]	Provides the definitive effect sizes for mood-congruent memory biases, proving positive baseline biases and negative depression filters, while discrediting mood-dependent memory.
McGaugh, J. L. (2000). *Science*.[13]	Proves that post-training emotional arousal

	activates the basolateral amygdala to modulate the consolidation and long-term stability of memories, dictating the encoding multiplier.
Eldar, E., & Niv, Y. (2015). *Nature Communications*.[25]	Provides a reinforcement learning framework where mood is shaped by reward prediction errors and subsequently biases expected values, inspiring the bidirectional coupling architecture.
Fleeson, W. (2001). *Journal of Personality and Social Psychology*.[32]	Establishes Whole Trait Theory, demonstrating that the vast majority of behavioral and state variance occurs within-person over time, providing the core variance validation target.
Scherer, K. R. (2005). *Social Science Information*.[20]	Outlines the Component Process Model, emphasizing that memory-dependent appraisal is the triggering mechanism for emotional episodes.
Open Spaced Repetition (2024). *GitHub Wiki*.[29]	Provides the exact closed-form math for power-law memory retrievability via the DSR (Difficulty, Stability, Retrievability) model, crucial for the lazy evaluation framework.
Rauthmann, J. F., et al. (2014). *Journal of Personality and Social Psychology*.[22]	Provides the categorization dimensions for situational construal (DIAMONDS), representing the interface between environmental events and internal appraisal.

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